

ERAP1 x axial spondyloarthritis

BOUNDED RECENT EVIDENCE ADDENDUM | 5 October 2026

The assessment incorporates a bounded addendum of recent primary studies from 2020-2026 for which sufficient methods and results were accessible for AXIS scientific curation.

CURRENT POSITION

Investigate context-specific ERAP1 modulation. Therapeutic benefit from ERAP1 inhibition is not established by this bounded evidence scope.

Decision: resolve the critical uncertainty first. No canonical DecisionState v2 has been created.

What the usable evidence changes

Recent accessible studies strengthen ERAP1-AS biological rationale, not validated therapeutic benefit. [CM01]

Full allotype, HLA/ERAP2 context and peptide substrate must be specified; a protective allotype is not universally inactive. [CM02]

Experimental benefit is endpoint-dependent: germline loss reduced rat arthritis incidence but worsened colon histology. [CM03]

AS is not every axSpA phenotype. Patient-genotype studies and disease models must not be silently generalized to every patient.

Additional potentially material literature was identified but not adjudicated where full methodological information was unavailable.

Independent scientific review pending.

Evidence strengthened. Attribution still open.

Recent accessible evidence	Decision relevance
Human genetics, PBMC and expanded CD8 responses	Stronger target/HLA context, not demonstrated drug benefit [CM01/05]
Substrate-sensitive recombinant and cellular allotype studies	Full sequence/context required; no universal low-activity label [CM02]
Divergent genetic-loss and inhibitor/KO phenotypes	Inhibition, lifelong loss and allosteric modulation differ [CM03/06]
New chemical series and regulatory-site complexes	More assay-specific tools; no intact-cell occupancy inferred [CM04]
Sparse ERAP2/IRAP counter-screen	Censored bounds retained; no non-comparable ratio [CM08]

Recent evidence addendum

10 primary studies integrated in the addendum; 21 atomic assertions and 10 impact assessments remain pending_review. This is not expert acceptance or a comprehensive review.

Four insufficient-access candidates remain outside assertion extraction. Corilagin 2025 is potentially material but its interaction/phenotype report cannot adjudicate intact-cell engagement without methods.

Clinical development, not efficacy

EAST1 (NCT07047703), an ERAP1-directed axSpA phase I/II trial, was registered with no posted results in the consulted record. No AS/axSpA efficacy results were identified in the searched sources. [REG01]

Chemical Learning: MODEL NOT BUILT in the unchanged indexed dataset. No new standardized measurements imported; recent-literature modelling eligibility was not assessed. [BASE01]

The next discriminating experiment

SAME PRIORITY, REFINED SPECIFICATION

Keep the matched-genotype engagement experiment; sharpen allotype, substrate, exposure and intracellular-versus-surface controls. [CM07]

Measure engagement and phenotype together

Use matched WT/null/rescue backgrounds, vehicle, dose/time, viability and interference controls. Resolve tested identity/form and validate a proximal cellular assay first. Specify HLA-B27 subtype, complete ERAP1 allotype and ERAP2 status. Distinguish intracellular folding/UPR from surface free-heavy-chain effects. [CM07]

Interpretable result	Scientific consequence
Engagement + dependency + rescue	Strengthens compound/context-specific ERAP1 attribution; translation and broader selectivity still need testing.
Phenotype persists without ERAP1	Weakens this compound's ERAP1 explanation, not the target universally.
Toxicity or invalid assay/genetic model	No mechanism conclusion; validate controls and repeat.

What remains unresolved

Relevant-exposure engagement and chemical-genetic dependency remain the discriminating gap for baseline phenotype-producing compounds. [CM06]

On-target, off-target, indirect and context-dependent explanations remain viable. New biology does not automatically select an inhibition strategy.

Bounded addendum does not replace the earlier failed exhaustive refresh: INSUFFICIENT REFRESH COMPLETENESS TO ASSESS. Full traceability and non-adjudicated reports are separated in the full brief.

Independent scientific review pending.

Recent Evidence 2020-2026

Integrated-study table

All ten entries: primary studies, sufficient inspected methods/results, INTEGRATE in this addendum only, pending independent review. Full experimental context, claim locators and limitations follow.

Study / year	Evidence layers	Accessibility / impact
wang-2022 / 2022	Genetics, Allotype/HLA, Mechanism, Patient-derived cellular	FULL_TEXT_ACCESSIBLE; DECISION_RELEVANT: Strengthens target biology and genotype-context rationale, not inhibition efficacy.
tran-2023 / 2023	Allotype/HLA, Mechanism, Genetic perturbation, Translational models	FULL_TEXT_ACCESSIBLE; DECISION_RELEVANT: Improves perturbational support while weakening a uniformly beneficial inhibition narrative.
tedeschi-2023 / 2023	Allotype/HLA, Mechanism, Patient-derived cellular	FULL_TEXT_ACCESSIBLE; DECISION_RELEVANT: Adds adaptive-immune context without establishing pathogenic causality or intervention direction.
hutchinson-2021 / 2021	Allotype/HLA, Mechanism, Biochemical pharmacology, Chemical perturbation	FULL_TEXT_ACCESSIBLE; DECISION_RELEVANT: Strengthens the requirement to specify allotype and peptide substrate rather than assign universal activity labels.
temponeras-2024 / 2024	Allotype/HLA, Mechanism, Biochemical pharmacology, Chemical perturbation	FULL_TEXT_ACCESSIBLE; DECISION_RELEVANT: Contradicts a universal low-activity interpretation of allotype 10, not the established association itself.
cyclohexyl-2024 / 2024	Chemical matter, Structures, Biochemical pharmacology, Chemical perturbation, Cellular phenotype, Selectivity	METHODS_AND_RESULTS_SUFFICIENT; DECISION_RELEVANT: Improves chemistry/structure options but does not resolve baseline compounds' cellular engagement.
maben-2021 / 2021	Mechanism, Structures, Biochemical pharmacology, Allotype/HLA	FULL_TEXT_ACCESSIBLE; DECISION_RELEVANT: Supports construct-specific mechanism without promoting structural behavior to occupancy in AS cells.
proteome-2025 / 2025	Mechanism, Immunopectidomics, Genetic perturbation, Chemical perturbation, Cellular phenotype	FULL_TEXT_ACCESSIBLE; DECISION_RELEVANT: Keeps indirect/context explanations viable and cautions against equating inhibition with loss of protein.
melanocyte-2025 / 2025	Allotype/HLA, Immunopectidomics, Genetic perturbation, Mechanism	FULL_TEXT_ACCESSIBLE; DECISION_RELEVANT: Supports allotype/HLA-specific measurements, not a universal functional knockout interpretation.
genomics-2026 / 2026	Genetics	FULL_TEXT_ACCESSIBLE; INFORMATIVE: Reinforces biology but neither validates inhibition nor demonstrates ERAP1-specific experimental causality.

Recent Evidence 2020-2026

Before / after

Categorical bounded interpretation, not numerical maturity scores. All inferences pending_review; canonical state unchanged.

Layer	Baseline	Recent / interpretation
Target-disease	Indexed target-AS biology, no validated therapy	Wang/Tran/genomics add human/model evidence; Biology strengthened; intervention validation separate
Genetics	Historical association	Taiwanese haplotypes and public European meta-analysis; Strengthened with ancestry and reused-cohort limitations
HLA interaction	B27-related phenotypes	B27-stratified human and B27/B7 model comparisons; Strengthened, not all HLA or all axSpA
Allotype context	Unresolved contextual uncertainty	Recombinant and expression-matched cellular studies; More important; labels and substrates cannot be interchanged
Mechanism	Partial peptide/FHC bridge	Folding, CD8 responses and broader cellular pathways; Richer but not one universally causal pathway
Perturbation	Genetic and chemical phenotypes	Divergent germline loss and inhibitor/KO effects; Loss, inhibition and allosteric modulation are not equivalent
Structure	Open 3QNF context	Conformational study plus four new regulatory-site complexes; More binding-site observations; not cellular occupancy or validated druggability
Chemistry	Three indexed Maben compounds	Cyclohexyl acid tool series; Additional options; no automatic baseline identity substitution

Recent Evidence 2020-2026

Before / after

Categorical bounded interpretation, not numerical maturity scores. All inferences pending_review; canonical state unchanged.

Layer	Baseline	Recent / interpretation
Biochemical pharmacology	Indexed context-fragmented activity	Allotype/substrate-sensitive assays; Stronger assay-specific evidence, no pooled universal potency
Cellular phenotype	Compound-treated engineered phenotypes	PBMC, CD8 and cancer-cell results; More context; patient phenotype is not clinical response
Cellular engagement	Not assessed in baseline	No integrated exposure-matched occupancy/proximal assay for baseline compounds; Not demonstrated by this addendum; inaccessible reports remain unresolved
Target dependency	Drug-specific gap	Genetic loss/re-expression and parallel chemical perturbation; Chemical-genetic dependency not demonstrated
Selectivity	Unresolved/non-comparable	Limited new-series ERAP2/IRAP counter-screen; Partially informative, NOT DIRECTLY COMPARABLE
Translation	No indexed disease rescue	Reduced rat arthritis incidence with divergent GI effects; Partial experimental phenotype modification, not human remission
Clinical	No efficacy indexed	EAST1 axSpA registration identified; no posted results; Development exists; no AS/axSpA efficacy established

Scientific questions

Q1-Q7

ID	Assessment	Rationale
Q1	PARTIALLY SUPPORTED	Recent accessible human association and rat perturbation strengthen target-AS biology, not validated therapeutic benefit.
Q2	SUPPORTED	Human stratification, CD8 subtype comparisons and B27/B7 model divergence strengthen HLA-context evidence.
Q3	SUPPORTED	Full allotype, HLA and substrate context matter; single SNPs and complete allotypes are not interchangeable.
Q4	NOT DEMONSTRATED	No universally preferred therapeutic direction established. Protective association, substrate-dependent activity and mixed organ outcomes do not choose inhibition for every patient.
Q5	SUPPORTED	New accessible recombinant pharmacology and cyclohexyl acid assays add biochemical evidence within specified contexts.
Q6	PARTIALLY SUPPORTED	Patient PBMC and expanded CD8 data add human cellular context; the seeded macrophage drug study remains abstract-only and is not adjudicated.
Q7	NOT DEMONSTRATED	Integrated structures and cellular phenotypes do not establish exposure-matched cellular engagement for the baseline phenotype-producing compounds.

Scientific questions

Q8-Q14

ID	Assessment	Rationale
Q8	NOT DEMONSTRATED	Parallel KO and inhibitor effects are not a matched inhibitor-in-null/rescue chemical-genetic dependency test.
Q9	PARTIALLY SUPPORTED	New-series counter-screen evidence is informative but sparse and cross-substrate; no selectivity ratio computed or baseline selectivity resolved.
Q10	PARTIALLY SUPPORTED	Germline ERAP1 loss modifies arthritis incidence in one experimental SpA model, with no improved GI clinical score and worse colon histology; not reversal of established disease.
Q11	NOT DEMONSTRATED	An ERAP1-directed axSpA trial registration was identified, but no AS/axSpA human interventional efficacy results were identified in the searched sources. This does not claim that no trial exists.
Q12	PARTIALLY SUPPORTED	REFINED wording/context; target engagement remains the critical uncertainty, not resolved or replaced.
Q13	PARTIALLY SUPPORTED	REFINED_EXPERIMENT: retain the chemical-genetic engagement design with more explicit allotype/substrate/compartment and exposure controls.
Q14	NOT ASSESSED	No new standardized learning measurements imported, so the indexed model refusal is unchanged. Eligibility of the entire recent chemical literature was not assessed.

Recent primary evidence wang-2022

Functional ERAP1 Variants Distinctively Associate with Ankylosing Spondylitis Susceptibility under the Influence of HLA-B27 in Taiwanese

DOI: 10.3390/cells11152427 | FULL_TEXT_ACCESSIBLE

INTEGRATE: addendum inclusion only. pending_review.

Design: human, case-control, patient-derived, genetic association, functional expression

Disease relevance: AS, revised New York criteria; not all axSpA

Experimental context

population: Taiwanese, same geographic region

cases: 863

controls: 1438

hla b27 positive cases: 801

discovery cDNA subjects: 45

functional PBMC donors: 7 ERAP1-001 homozygotes versus 4 ERAP1-002 homozygotes, all AS/HLA-B*27:04

allotype: study-local ERAP1-001 and ERAP1-002 labels; not automatically Hap1/Hap2

001 residues: 56E/127R/276I/349M/528K/575D/725R/730Q

002 residues: 56E/127P/276M/349M/528R/575D/725R/730E

assays: Sanger full-length cDNA; TaqMan SNVs; PBMC PMA 10 ng/mL 48h then LPS 1 microgram/mL 16h; CHO/U937 expression; immunoprecipitated enzyme fluorogenic assay

treatment context: not sufficiently described for confounding adjustment

Atomic source assertions (not independent expert approval)

Study-local ERAP1-001 was enriched among AS cases versus controls (adjusted OR 1.53, 95% CI 1.35-1.73).

addendum:claim:wang-2022:1 | Results 3.3/Table 1

In HLA-B27-negative participants, ERAP1-002 was associated with AS versus HLA-B27-negative controls (adjusted OR 1.69, 95% CI 1.15-2.49).

addendum:claim:wang-2022:2 | Results 3.4/Table 3

The small HLA-B*27:04 AS PBMC comparison linked ERAP1-001 to higher ERAP1, FHC/UPR markers and IL-23 than ERAP1-002; CHO expression also differed.

addendum:claim:wang-2022:3 | Results 3.5/Figures 1-2; Methods 2.9/2.13-2.15

Evidence impact / limitations

strengthens; DECISION_RELEVANT: Strengthens target biology and genotype-context rationale, not inhibition efficacy.

Sex imbalance: 717/863 male cases versus 706/1438 controls; sex-adjusted association; Small functional donor subset and technical replicates; HLA-B27-positive case/control haplotype comparison not significant; Methods describe fluorogenic conditions but do not unambiguously identify the substrate sequence; do not manufacture it; Discussion-like inference that 002 is risk in B27-negative is supported by stratified controls, not just case-case comparison; No pharmacological direction or engagement test

Recent primary evidence tran-2023

Paradoxical Effects of Endoplasmic Reticulum Aminopeptidase 1 Deficiency on HLA-B27 and its Role as an Epistatic Modifier in Experimental Spondyloarthritis

DOI: 10.1002/art.42327 | FULL_TEXT_ACCESSIBLE

INTEGRATE: addendum inclusion only. pending_review.

Design: animal, genetic perturbation, prospective phenotype monitoring

Disease relevance: HLA-B27 transgenic experimental SpA; not human axial disease remission

Experimental context

model: 33-3 HLA-B*27:05/human beta2-microglobulin transgenic rats, mixed Lewis/Sprague-Dawley

controls: HLA-B7 transgenic and non-transgenic, ERAP1 +/+, +/-, -/-

intervention: TALEN germline ERAP1 exon-1 deletion

cells: bone-marrow-derived macrophages and dendritic cells

phenotype: peripheral arthritis, epididymoorchitis, stool and colon histology

blinding: clinical caretakers and histology readers blinded

randomization: not stated in main methods

Atomic source assertions (not independent expert approval)

Arthritis frequency was 12/37 with ERAP1 versus 4/34 without ERAP1 ($p=0.03$); severity and onset among affected animals were comparable.

addendum:claim:tran-2023:1 | Results/Figure 4A

ERAP1 loss did not improve stool scores and increased colon histology scores.

addendum:claim:tran-2023:2 | Results/Figure 4C

ERAP1 loss reduced intracellular HLA-B27 misfolding/UPR but increased surface free heavy chains and dimers.

addendum:claim:tran-2023:3 | Results/Figures 1-3

ERAP1 loss reduced excess IL23a induction but did not restore defective dendritic-cell stimulation or the depleted CD103-positive population.

addendum:claim:tran-2023:4 | Results/Figure 5

Evidence impact / limitations

opens_new_uncertainty; DECISION_RELEVANT: Improves perturbational support while weakening a uniformly beneficial inhibition narrative.

Germline prevention is not reversal by a drug; Multiple transgene copies and rodents without ERAP2; No human clinical efficacy; Compartment-specific FHC effects; No single disease-wide beneficial direction

Recent primary evidence tedeschi-2023

ERAP1 and ERAP2 Haplotypes Influence Suboptimal HLA-B*27:05-Restricted Anti-Viral CD8+ T Cell Responses Cross-Reactive to Self-Epitopes

DOI: 10.3390/ijms241713335 | FULL_TEXT_ACCESSIBLE

INTEGRATE: addendum inclusion only. pending_review.

Design: human, haplotype association, patient-derived, cell, computational

Disease relevance: AS revised New York criteria; mechanistic adaptive immune context, not therapy

Experimental context

population: Italian recruitment, ancestry not explicitly established

hla: 102 B*27:05 subjects (93 AS/9 healthy); 11 B*27:09 controls; 13 B*07 carriers

genotype: four-marker ERAP1/ERAP2 haplotype, not full ERAP1 protein allotype

markers: ["rs27044", "rs30187", "rs75862629", "rs2248374"]

readout: IFN-gamma-positive CD8 after 14-day peptide/IL-2 expansion

peptides: pEBNA3A RPPIFIRRL, pDYNEIN RPPIFGDFL, RPXXXXXXL cocktail

dose: pEBNA3A 20 microgram/mL; pDYNEIN 40 microgram/mL initial culture

controls: APC medium subtraction, EBV serology, alternative HLA subtypes, T2 stabilization

treatment context: not controlled in reported analysis

Atomic source assertions (not independent expert approval)

The four-marker CCAG ERAP1/ERAP2 haplotype was enriched in pEBNA3A nonresponders among B*27:05 carriers (73 responders/29 nonresponders; haplotype $p=0.0179$).

addendum:claim:tedeschi-2023:1 | Results 2.1/Tables 1-2; Methods 4.1/4.2/4.9

Expanded B*27:05-restricted pEBNA3A responses could cross-react with the dynein self peptide; responses also occurred in healthy donors.

addendum:claim:tedeschi-2023:2 | Results 2.3/Figure 4; Methods 4.6-4.8

Evidence impact / limitations

strengthens; DECISION_RELEVANT: Adds adaptive-immune context without establishing pathogenic causality or intervention direction.

Long ex-vivo expansion and arbitrary responder threshold; Healthy-donor cross-reactivity limits disease specificity; Dynein residue numbering differs between Results and Methods; retain peptide sequence without asserting a resolved coordinate; Not direct immunopeptidome mass spectrometry; No engagement or chemical-genetic test

Recent primary evidence hutchinson-2021

Common allotypes of ER aminopeptidase 1 have substrate-dependent and highly variable enzymatic properties

DOI: 10.1016/j.jbc.2021.100443 | FULL_TEXT_ACCESSIBLE

INTEGRATE: addendum inclusion only. pending_review.

Design: recombinant, biochemical, population haplotype estimation

Disease relevance: Contextual ERAP1 biology; not AS patient intervention

Experimental context

population: 2504 phased 1000 Genomes samples, five major ancestry groups

allotype: ten recombinant full-length common allotypes, noncleavable C-terminal 6-His tag

substrates: ["Leu-AMC", "Leu-pNA", "LLRIQRGPGRAFVTI", "LLKHHAFSFK", "YTAFTIPSI", "GLEQLESIIINFEKL"]

compounds: DG013A active-site inhibitor; GSK849 regulatory-site modulator

assays: fluorescence/absorbance, HPLC and mass-spectrometry kinetics

replication: peptide trimming triplicates; allotype-10 enzyme concentration increased where specified

Atomic source assertions (not independent expert approval)

Ten common recombinant ERAP1 allotypes differed in substrate-dependent trimming and intermediate accumulation; YTAFTIPSI catalytic efficiency differed up to 60-fold.

addendum:claim:hutchinson-2021:1 | Results/Figures 2-3; Experimental procedures

GSK849 activated Leu-AMC hydrolysis but inhibited YTAFTIPSI trimming, with allotype-dependent potency; DG013A inhibited all ten allotypes in the Leu-AMC assay.

addendum:claim:hutchinson-2021:2 | Results/Figure 4; Experimental procedures enzymatic assays

Evidence impact / limitations

contextualizes; DECISION_RELEVANT: Strengthens the requirement to specify allotype and peptide substrate rather than assign universal activity labels.

Activity sums assume independent alleles and do not model expression differences; Substrate-specific rank cannot be generalized to every peptide; No AS cellular engagement or clinical evidence

Recent primary evidence temponeras-2024

Polymorphic positions 349 and 725 of the autoimmunity-protective allotype 10 of ER aminopeptidase 1 are key in determining its unique enzymatic properties

DOI: 10.3389/fimmu.2024.1415964 | FULL_TEXT_ACCESSIBLE

INTEGRATE: addendum inclusion only. pending_review.

Design: recombinant, biochemical, site-directed mutagenesis

Disease relevance: Contextual allotype mechanism, not disease rescue

Experimental context

allotype: allotypes 1/10 and allotype-10 V349M, Q725R, double mutant; insect expression with C-terminal TEV/10-His construct

substrates: ["YTAFTIPSI", "LRVYEKMALY", "VSVRSRRCLRL", "Leu-AMC", "Leu-pNA"]

compounds: DG013A and GSK849; source contains inconsistent GSK labels in some captions

conditions: peptide 10 micromolar, enzyme concentration adjusted by construct; 1h 37C triplicates; GSK849 10 micromolar peptide experiment

controls: wild-type allotypes, verified mutations, solvent controls

Atomic source assertions (not independent expert approval)

Allotype 10 was much less active than allotype 1 for YTAFTIPSI but more active for LRVYEKMALY; V349M/Q725R effects also depended on substrate.

addendum:claim:temponeras-2024:1 | Results/Figure 2; peptide-trimming methods

Allotypes 1 and 10 showed different substrate-concentration kinetics and DG013A inhibition patterns; GSK849 modulation also depended on polymorphism and substrate.

addendum:claim:temponeras-2024:2 | Results/Figures 3-5; Michaelis-Menten and inhibitor methods

Evidence impact / limitations

contradicts; DECISION_RELEVANT: Contradicts a universal low-activity interpretation of allotype 10, not the established association itself.

Not an AS cohort; Substrate concentration and kinetic model affect interpretation; Compound label discrepancies retained; no new exact chemical identity asserted

Recent primary evidence cyclohexyl-2024

Optimization

of Potent and Selective Cyclohexyl Acid

ERAP1 Inhibitors Using Structure- and Property-Based Drug Design

DOI: 10.1021/acsmchemlett.4c00401 | METHODS_AND_RESULTS_SUFFICIENT

INTEGRATE: addendum inclusion only. pending_review.

Design: biochemical, structural, cell, animal PK

Disease relevance: Tool pharmacology, HeLa antigen presentation and rat PK; not AS efficacy

Experimental context

construct: paper reports Hap2 for biochemical assay; PDB construct mapping requires separate reconciliation

substrate: YTAFTIPSI to TAFTIPSI, high-throughput MS

cell: HeLa flow-cytometry SIINFEKL presentation, not ERAP1 occupancy

compounds: paper-local cyclohexyl acids 1, 2, 7, 13; structures depicted in source and bound in linked PDB

replication: main tables $n \geq 3$ unless stated; ERAP2/IRAP counter-screen of 7 $n=1$

selectivity substrates: ERAP2 Arg-AMC, IRAP Leu-AMC

chemical identity: paper/PDB ligand bridge only, tested-form/stereochemistry not promoted to verified registry substance

supporting methods: main methods sufficient for listed qualitative assertions; SI not inspected in this addendum

Atomic source assertions (not independent expert approval)

Optimization improved peptide-trimming inhibition and HeLa SIINFEKL-presentation activity; hit 1 lacked measurable cellular activity despite biochemical activity.

addendum:claim:cyclohexyl-2024:1 | Main text assay description and Tables 1-2

Compound 7 had reported ERAP2 and IRAP $pIC_{50} < 4.0$ with $n=1$, using Arg-AMC and Leu-AMC respectively; these are not directly comparable to ERAP1 peptide-substrate potency.

addendum:claim:cyclohexyl-2024:2 | Main selectivity paragraph

Ligand-bound ERAP1 structures were reported for compounds 1, 2, 7 and 13 (9GJN/9GK6/9GJS/9GKE), deposited in 2024 and released in January 2025.

addendum:claim:cyclohexyl-2024:3 | Figures 2/4/5; Data availability; RCSB entry records

Evidence impact / limitations

fills_gap; DECISION_RELEVANT: Improves chemistry/structure options but does not resolve baseline compounds' cellular engagement.

Detailed SI and form/structure standardization required before quantitative learning import; Different substrates/constructs and biochemical versus cellular endpoints; No intact-cell engagement or genetic-dependency readout; GSK author employment conflicts disclosed

Recent primary evidence maben-2021

Conformational dynamics linked to domain closure and substrate binding explain the ERAP1 allosteric regulation mechanism

DOI: 10.1038/s41467-021-25564-w | FULL_TEXT_ACCESSIBLE

INTEGRATE: addendum inclusion only. pending_review.

Design: recombinant, structural, biochemical, computational

Disease relevance: Contextual ERAP1 conformational mechanism; not AS efficacy

Experimental context

variant: matched ERAP1 Lys528/Arg528 constructs

assays: SAXS/WAXS, crystal structures, Leu-AMC/Leu-pNA and peptide assays

compound: study compounds 2/3 linked to historical Maben chemistry; not cyclohexyl compound 3

replication: two independent activity experiments with triplicates; SAXS independent-experiment limits reported

controls: apo protein, open/closed structural models and distinct ligand classes

Atomic source assertions (not independent expert approval)

The allosteric modulator 3 induced ERAP1 closure in solution; Lys528/Arg528 altered potency for closure-inducing DG013 and modulator 3 but not inhibitor 2.

addendum:claim:maben-2021:1 | Results Figures 4-5; SAXS/Activity methods

Evidence impact / limitations

contextualizes; DECISION_RELEVANT: Supports construct-specific mechanism without promoting structural behavior to occupancy in AS cells.

Preferred compound docking pose is not a measured binding pose; Conformation under saturating purified-protein conditions is not cellular engagement; 2023 correction adds facility acknowledgement, not changed scientific results

Recent primary evidence proteome-2025

ERAP1 Activity Modulates the Immunopeptidome but Also Affects the Proteome, Metabolism, and Stress Responses in Cancer Cells

DOI: 10.1016/j.mcpro.2025.100964 | FULL_TEXT_ACCESSIBLE

INTEGRATE: addendum inclusion only. pending_review.

Design: cell, genetic perturbation, chemical perturbation, proteomics

Disease relevance: A375 melanoma and THP-1 leukemia; contextual, not AS patient evidence

Experimental context

compound: source compound 3: 4-methoxy-3-(N-(2-(piperidin-1-yl)-5-(trifluoromethyl)phenyl)sulfamoyl)benzoic acid; not Maben compound 3

dose: 10 micromolar, 0.1% DMSO, 6 days for omics; 48h downstream assays

hla: A375 A*01:01/A*02:01/B*44:03/B*57:01/C*06:02/C*16:01, not B27

ERAP2: A375 low/absent expression genotype

replication: 2 biological immunopeptidome replicates/3 technical; 4 A375 and 3 THP-1 proteome biological replicates

control: KO, WT, vehicle; no dose-matched inhibitor-in-KO epistasis/rescue experiment

Atomic source assertions (not independent expert approval)

Chemical inhibition and ERAP1 KO produced distinct A375 immunopeptidomes despite both perturbing peptide presentation.

addendum:claim:proteome-2025:1 | Results Figures 1-2; Experimental procedures

Proteomic and ROS changes accompanied ERAP1 disruption; shared effects do not isolate direct peptide processing from indirect cellular pathways.

addendum:claim:proteome-2025:2 | Results Figures 4-5; Experimental procedures

Evidence impact / limitations

opens_new_uncertainty; DECISION_RELEVANT: Keeps indirect/context explanations viable and cautions against equating inhibition with loss of protein.

A375 KO 1G5 omics clone had partial genomic duplication; downstream experiments switched to 1B12; Healthy-donor cytotoxicity assay not patient-derived disease efficacy; KO/inhibitor concordance is not demonstrated compound dependency; Technical images/replicates are not additional donors

Recent primary evidence melanocyte-2025

ERAP1 Allotypes 2 and 10 Differentially Regulate the Immuno-peptidome of Melanocytes

DOI: 10.1111/imm.70056 | FULL_TEXT_ACCESSIBLE

INTEGRATE: addendum inclusion only. pending_review.

Design: cell, genetic perturbation, allotype re-expression, mass spectrometry

Disease relevance: Melanoma/psoriasis-context HLA-I biology, not direct AS/HLA-B27 evidence

Experimental context

model: A375 KO clone 1B12 reconstituted with single ERAP1 allotype 2 or 10

controls: selected clones with comparable ERAP1 expression

dose: IFN-gamma 20 ng/mL 24h

replication: 3 separate biological cultures, 2 DIA technical runs plus 1 DDA per sample; one selected clone per allotype

hla: A375 HLA-A/B/C, not HLA-B27

analysis: W6/32 purification, LC-MS/MS, 1% FDR, differential $\log_2 \geq 1/q \leq 0.05$

sequence: full construct sequences source Figure S1, not imported as verified protein registry identities

Atomic source assertions (not independent expert approval)

Expression-matched allotype-2 and allotype-10 A375 clones differed mainly in peptide abundance, also in length and predicted HLA assignment; allotype 10 retained cellular trimming activity.

addendum:claim:melanocyte-2025:1 | Results 3.2-3.4/Figures 2-5; Methods 2.2-2.6

Evidence impact / limitations

contextualizes; DECISION_RELEVANT: Supports allotype/HLA-specific measurements, not a universal functional knockout interpretation.

Clone effects not excluded by multiple independent clones per allotype; Predicted HLA affinities are not measured binding; No normalization to a healthy immuno-peptidome and no therapeutic benefit endpoint

Recent primary evidence genomics-2026

Multi-layered functional genomics prioritizes candidate effectors and regulatory mechanisms of ankylosing spondylitis

DOI: 10.3389/fimmu.2026.1859310 | FULL_TEXT_ACCESSIBLE

INTEGRATE: addendum inclusion only. pending_review.

Design: human, GWAS meta-analysis, computational fine-mapping, cTWAS

Disease relevance: AS public GWAS, not a newly independent patient cohort

Experimental context

population: European ancestry MVP, FinnGen R12, UKB-WGS

cases: 7551

controls: 1258581

methods: allele-harmonized inverse-variance fixed/random effects, cohort sensitivity, fine-mapping/cTWAS

hla: B*27 dosage unavailable, no conditional B*27 analysis

endpoint: AS susceptibility, not disease response to intervention

Atomic source assertions (not independent expert approval)

The public-cohort meta-analysis recapitulated the ERAP1 susceptibility locus ($p=5.20e-21$), with rs26509 prioritized by three of four fine-mapping methods.

addendum:claim:genomics-2026:1 | Results GWAS/Fine-mapping; Methods study populations/QC

Evidence impact / limitations

strengthens; INFORMATIVE: Reinforces biology but neither validates inhibition nor demonstrates ERAP1-specific experimental causality.

Reuses public cohorts; not an independent replication of studies using the same biobanks; Substantial heterogeneity and cohort sensitivity; ERAP1 not among seven listed cTWAS-colocalization convergent genes; No experimental ERAP1 knockdown; Jurkat follow-up concerns other genes; AS phenotype ascertainment relies on public GWAS datasets, not harmonized clinical recruitment

Potentially Material - Not Fully Adjudicated

Additional potentially material literature was identified but not adjudicated where full methodological information was unavailable.

These are access/adjudication records, NOT accepted source assertions. No conclusion from them resolves engagement or chooses therapeutic direction.

babaie-2023 | ABSTRACT_ONLY

DOI: 10.1016/j.clim.2023.109268

Public canonical TDM response contains metadata only. Cannot verify compound identity, dose, exposure, differentiation protocol, readout controls or engagement/dependency.

Reported, not adjudicated: Abstract reports an AS macrophage FHC/NK-cell effect after ERAP1 inhibition; no new source assertions extracted.

wu-mr-2025 | ABSTRACT_ONLY

DOI: 10.1016/j.lfs.2025.123682

OpenAlex now labels the paper hybrid OA, but the public TDM response still has no scientific body and DOI/publisher reading failed. OA metadata is not sufficient methods access.

Reported, not adjudicated: Abstract: five proteomic GWAS, FinnGen R10 3162 cases/294770 controls, UKB replication 1344/324074, European ancestry; genetically proxied circulating ERAP1 positively associated with AS. Instrument identity/strength, exact MR estimator and pleiotropy details not inspected. Abstract lists coloc support for IL23R/IL1RL2, not ERAP1. MR is not drug inhibition validation.

corilagin-2025 | ABSTRACT_ONLY

DOI: 10.1016/j.intimp.2025.115180

Publisher 403; public TDM response metadata only; no repository full text identified. Abstract-level claims cannot establish intact-cell versus lysate/protein interaction, exposure matching, selectivity, dependency or controls.

Reported, not adjudicated: Accessible report describes compound screening, ERAP1 interaction at protein/cellular levels, enzyme inhibition/active-site rationale and HLA-B27-related cellular effects. These are reported findings only, not accepted AXIS assertions or a resolved engagement gap.

carnosic-2024 | SUPPLEMENT_ONLY

DOI: 10.1021/acs.jafc.4c00957

Main text re-check failed; public SI previously inspected in historical refresh, provenance commit 97c546e. Supplement does not establish complete methods, chemical-label bridge or cellular engagement. No addendum source assertions or measurements.

Reported, not adjudicated: No new assertions extracted.

Selection boundaries & provenance

18 selected candidates from 176 retrieved metadata records. Eight bounded searches and nine seed-verification queries do not constitute complete screening. Default ordering/truncation, accessibility and purposeful selection can bias this sample. 10 integrate; 1 contextualize; 1 already indexed; 4 insufficient access; 2 not integrated.

INTEGRATE studies are detailed individually above. Complete selected-study table and rationale are in the frozen study-selection.md/candidate-studies.json.

Other selected records

maben-indexed-2020: ALREADY_INDEXED. Already part of baseline pharmacology/cellular/Phase 3.9 packages; do not count as newly integrated evidence. Earliest online date in 2019; 2020 print label is not new 2020 discovery.

benzofuran-2021: DO_NOT_INTEGRATE. PMC indexed, but XML read returned front matter without body. Not selected for detailed publisher/SI curation within this bounded attempt; no claim that the paper is intrinsically inaccessible or weak.

immunopeptidome-2023: DO_NOT_INTEGRATE. Publisher full text and institutional PDF located. Detailed methods/SI not fully inspected here; later 2025 A375 study selected instead, without claiming independent replication or extracting this paper's claims.

east1-registry: CONTEXTUALIZE. Accessible primary registry supports development context, not efficacy. Recruiting; hasResults false at retrieval. Sponsor-reported GRWD0715 identity not standardized chemical structure. Registration is not an intervention result.

Inference and source are separate

AI-assisted source extraction and recommendation wording remain pending independent scientific review. Deterministic integrity/boundary checks do not approve scientific claims or independently validate an experiment.

traceability.json maps each CM conclusion to study-level AXIS inference, atomic claim, experiment/context, DOI and section locator. REG01 is a registration fact with primary registry provenance, not a clinical source assertion. BASE01 is unchanged baseline learning status.

Original v1 brief and canonical resources are unchanged. This bounded addendum does not supersede INSUFFICIENT REFRESH COMPLETENESS TO ASSESS.

Independent scientific review pending.